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23/3/26 U Parvathi

Q) solve the 0/1 knapsack optimization problem by applying branch & bound technique, explore feasible & non-feasible solution Compute the upper bounds and determine the optimal set of items that maximizes profit without exceeding knapsack capacity.

weights [2, 3, 1, 5] profits [10, 15, 7, 8]

knapsack capacity m=5 using Fifo & LC

Sol:

Profit=0, weight=0

$UB = 7+10+(15 \times \frac{2}{3})$

$= 7+10+10 = 27$

Node 1: Include item 3

wt=1, pf=7

UB = 27

Node 2: Include item 3

wt=3, pf=17

UB=27

Node 3: Include item 2

wt=6 (not feasible)

Node 5: Include item 4

wt = 8 [Not feasible]

Node 4: Exclude item 2

Node 6: Exclude item 4

wt=3, pf=17

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Other combinations

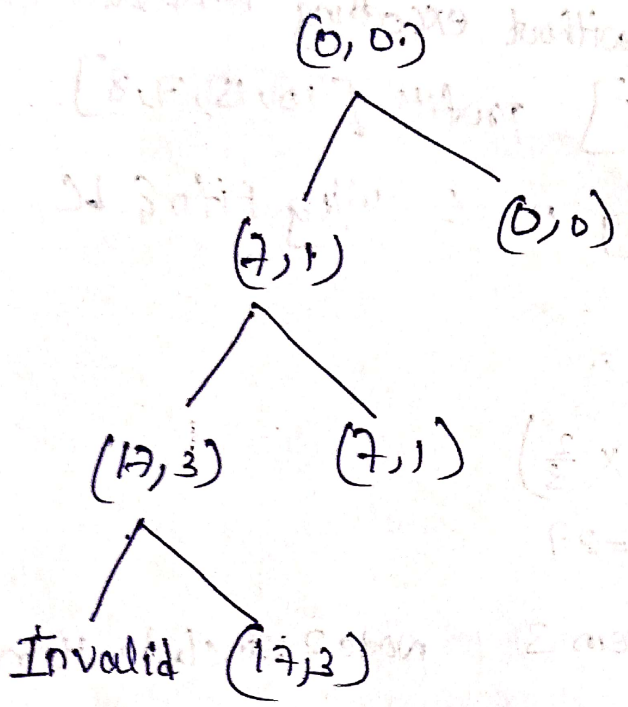
Item(3+2) ; Item(4+2) ; Item(3+1+2)

Pf = 22 | Pf = 25 | not feasible

wt = 4 | wt = 5

∴ Optimal profit = 25

State Space tree



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ALM-8

3/26

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1) Apply dynamic programming for 0/1 knapsack where $m=7$, $n=4$, $w=[1,3,4,5]$, $P=[1,4,5,7]$

2) Using least cost branch & bound solve the following knapsack problem $m=8$, $n=4$, $w=\{2,4,6,9\}$, $P=\{10,10,12,18\}$

2A) $8-2=6$, $6-4=2$, $8-6=2$

Profit $= P_1+P_2+P_3 = 10+10+12 = \$20$

So, upperbound $= \$20$

As branch & bound is a minimum problem we consider negative profits

$\therefore$ Upperbound $= -\$20$

Lower bound (with fraction) $= 10+10+\left(\frac{2}{6}\times 12\right) = -24$

Steps: For node 2, $x_1=1$

$U = 10+10 = -20$

$L = -24$

For $x_1=0$, node 8

$U = 10 \rightarrow -10$

$L = 10+\frac{4}{6}\times 12 = 10+8 = -18$

For node 2 $\Rightarrow U-L=4$

For node 3 $\Rightarrow U-L=8$

$\therefore$ we expand node 2

For node 4: $x_1=1$, $x_2=1$

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$U = 10+10 = -20$

$L = 10+10+4 = -24$

For node 5: $x_1 = 1$, $x_2 = 0$

$U = 10+12 = -22$

$L = 10+12 = -22$

for node 4 $\Rightarrow U-L = 4$

for node 5 $\Rightarrow U-L = 0$

we expand node 4, since for node 5 there is no need of branching.

For node 6 : $x_1 = 1$, $x_2 = 1$, $x_3 = 1$

$wt = 2+4+6 = 12 > 8$ (not feasible)

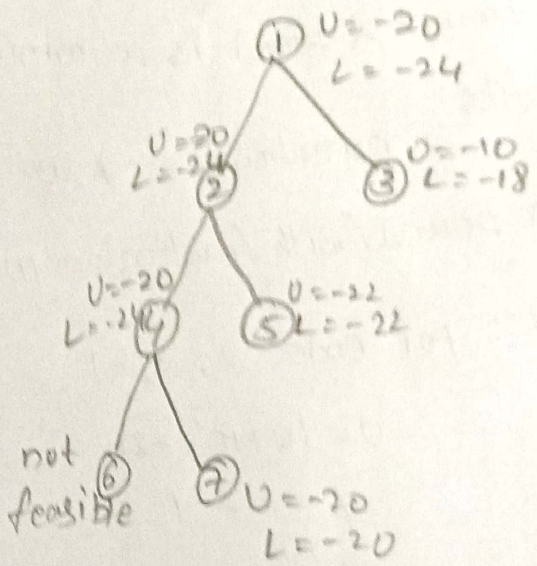
for node 7: $x_1 = 1$, $x_2 = 1$, $x_3 = 0$

$wt = 6$

$U = 10+10 = -20$

$L = 20 = -20$

Tree:



∴ Final Answer

Item selected: Item 1 & item 2

Total weight = 8

Maximium profit = 20

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HA)

Given

$m = 7, n = 4$

$W = \{1, 3, 4, 5\}$

$P = \{1, 4, 5, 7\}$

i/j	0	1	2	3	4	5	6	7
0	0	0	0	0	0	0	0	0
1	0	1	1	1	1	1	1	1
2	0	1	1	4	5	5	5	5
3	0	1	1	4	5	6	6	9
4	0	1	1	4	5	7	8	9

$$V[i][j] = \begin{cases} 0 \\ V[i-1][j] \\ \max(V[i-1][j],\ P_i + V[i-1]) \end{cases}$$

$V[4][7] = 9$

Backtracking from $V[4][7] = 9$

Compare with $V[3][7] = 9 \rightarrow$ item 4 not include

$V[3][7] \neq V[2][7] \rightarrow$ item 3 include

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remaining capacity = 7 − 4 = 3

v[2][3] ≠ v[1][3] → item 2 include

Max profit = 9, selected items = {item2, item3}

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